# MCS3301 – Control Engineering I

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# Mathematical System Models

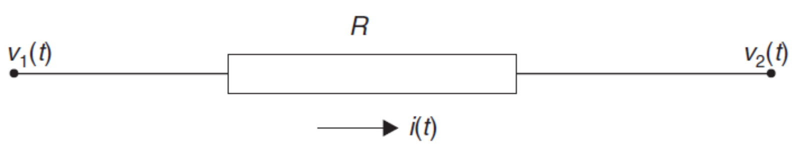
- Mathematical models of physical systems are key elements in the design and analysis of control systems.
- To understand and control complex systems, one must obtain quantitative mathematical models of these systems.
- It is necessary therefore to analyze the relationships between the system variables and to obtain a mathematical model.
- This can be done from the knowledge of the physical characteristics of the system.
- Alternatively, it can also be done via experimentation i.e. by measuring how the system's outputs responds to known inputs.
- Because the systems under consideration are dynamic in nature, the descriptive equations are usually differential equations.

# Linear Differential Equations

- An $n^{th}$ order linear differential equation takes the form

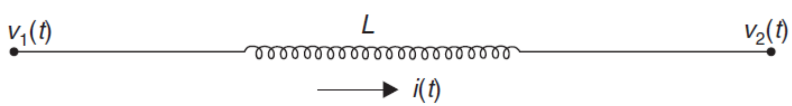
$$\frac{d^n y(t)}{dt^n} + a_{n-1}\frac{d^{n-1} y(t)}{dt^{n-1}} + \cdots + a_o y(t) = b_m \frac{d^m u(t)}{dt^m} + b_{m-1}\frac{d^{m-1} u(t)}{dt^{m-1}} + \cdots + b_o u(t)$$

- Example: Electrical circuits can be modelled based on the following relationships



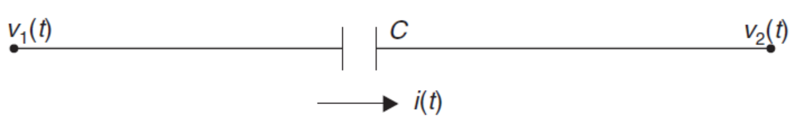
(a) Resistance

$$v_1(t) - v_2(t) = Ri(t)$$



(b) Inductance

$$v_1(t) - v_2(t) = L\frac{di}{dt}$$

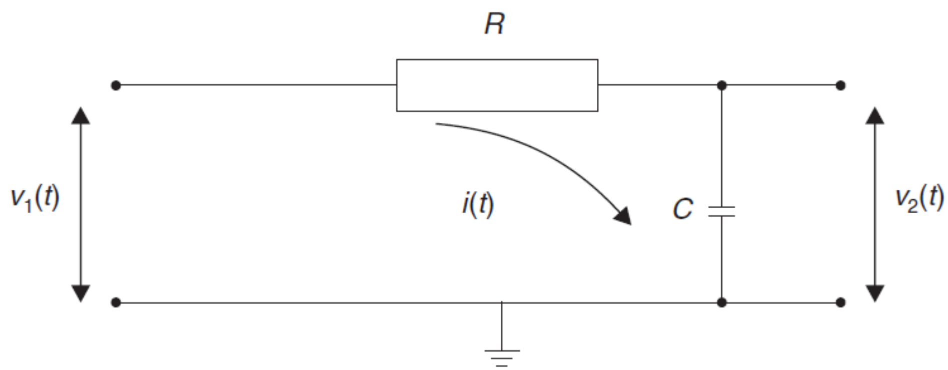


(c) Capacitance

$$v_1(t) - v_2(t) = \frac{1}{C}\int i\, dt$$

# Mathematical Modelling: RC Example

- Consider the RC circuit below




$$v_1(t) - v_2(t) = Ri(t)$$

$$v_2(t) = \frac{1}{C}\int i\,dt$$

OR

$$C\frac{dv_2}{dt} = i(t)$$

Thus

$$v_1(t) - v_2(t) = RC\frac{dv_2}{dt}$$

$$RC\frac{dv_2}{dt} + v_2 = v_1(t)$$

- Thus the system can be modelled as a first order differential equation.

# Mechanical Systems

- An elastic (spring) element is assumed to extend proportionally to the force applied to it

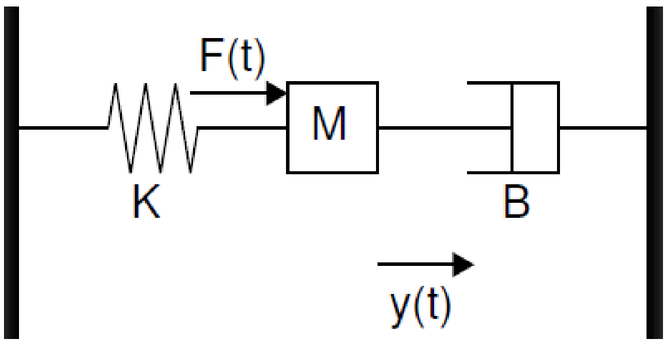
$$F = Ky$$

- A damping element produces a velocity proportional to the force applied to it

$$F(t) = Bv(t) = B\frac{dy}{dt}$$

- A body is accelerated by the product of its mass and acceleration

$$F(t) = Ma(t) = M\frac{dv}{dt} = M\frac{d^2y}{dt^2}$$




$$M\frac{d^2y}{dt^2} + B\frac{dy}{dt} + Ky = F(t)$$

$$\frac{d^2y}{dt^2}(t) + \frac{B}{M}\frac{dy}{dt}(t) + \frac{K}{M}y(t) = \frac{1}{M}F(t)$$

# Laplace Transforms

- To compute the time response of a dynamic system, its differential equation i.e. the system model has to be solved.
- However, differential equations can be quite complex to solve.
- The Laplace transform transforms the problem from the time $(t)$ domain to the Laplace $(s)$ domain.
- Thus complex time domain DEs become relatively simple algebraic equations.
- The Laplace transform of a function $f(t)$ is defined by the integral

$$L[f(t)] = \int_0^\infty f(t)e^{-st}dt = F(s)$$

where $s$ is the complex entity $\sigma \pm j\omega$

# Laplace Transforms: Common Functions

- Example 1. $f(t) = 1$ is a unit step function defined as

$$f(t) = \begin{cases} 1, & t > 0 \\ 0, & otherwise \end{cases}$$

$$L[f(t)] = F(s) = \int_0^\infty 1e^{-st}dt$$

$$= \left[-\frac{1}{s}(e^{-st})\right]_0^\infty$$

$$= \left[-\frac{1}{s}(0-1)\right] = \frac{1}{s}$$

- Example 2. Exponential function: $f(t) = e^{-at}$

$$L[f(t)] = F(s) = \int_0^\infty e^{-at}e^{-st}dt$$

$$= \int_0^\infty e^{-(s+a)t}dt$$

$$= \left[-\frac{1}{s+a}(e^{-(s+a)t})\right]_0^\infty$$

$$= \left[-\frac{1}{s+a}(0-1)\right]$$

$$= \frac{1}{s+a}$$

# Laplace Transforms: Common Functions

- Example 3. Dirac delta function

  The Dirac delta function $\delta(t)$ is an impulse of infinite magnitude infinitesimal duration and unit area. It has the fundamental property

$$\int_{-\infty}^{\infty} f(t)\delta(t)dt = f(0) \text{ for any } f(t)$$

Its Laplace transform is thus

$$L[\delta(t)] = \int_0^{\infty} \delta e^{-st} dt$$
$$= e^{-s\times 0}$$
$$= 1$$

- Laplace transforms of common functions can be found in Laplace transform tables.

# Laplace Transform Table

| | *Time function* $f(t)$ | *Laplace transform* $\mathscr{L}[f(t)] = F(s)$ |
|---|---|---|
| 1 | unit impulse $\delta(t)$ | $1$ |
| 2 | unit step 1 | $1/s$ |
| 3 | unit ramp $t$ | $1/s^2$ |
| 4 | $t^n$ | $\dfrac{n!}{s^{n+1}}$ |
| 5 | $e^{-at}$ | $\dfrac{1}{(s+a)}$ |
| 6 | $1 - e^{-at}$ | $\dfrac{a}{s(s+a)}$ |
| 7 | $\sin \omega t$ | $\dfrac{\omega}{s^2+\omega^2}$ |
| 8 | $\cos \omega t$ | $\dfrac{s}{s^2+\omega^2}$ |
| 9 | $e^{-at} \sin \omega t$ | $\dfrac{\omega}{(s+a)^2+\omega^2}$ |
| 10 | $e^{-at}(\cos \omega t - \dfrac{a}{\omega} \sin \omega t)$ | $\dfrac{s}{(s+a)^2+\omega^2}$ |

# Properties of the Laplace Transform

1. Differentiation:

$$L\left[\frac{d^n}{dt^n} f(t)\right] = s^n F(s) - f(0)s^{n-1} - f'(0)s^{n-2} - \ \ldots$$

Where $f(0)$ and $f'(0)$ are initial conditions. For example the Laplace transform of $M\frac{d^2y}{dt^2} + B\frac{dy}{dt} + Ky = F(t)$ is

$$M\left(s^2Y(s) - sy(0) - \frac{dy}{dt}(0)\right) + B\big(sY(s) - y(0)\big) + KY(s) = F(s)$$

2. Linearity

$$L[f_1(t) + f_2(t)\ ] = \ F_1(s) + F_2(s)$$

3. Initial value theorem

$$f(0) = \ \lim_{t\to 0}[f(t)] = \ \lim_{s\to\infty}[sF(s)]$$

4. Final value theorem

$$f(\infty) = \ \lim_{t\to\infty}[f(t)] = \ \lim_{s\to 0}[sF(s)]$$

# Laplace Transform: Example

- Given zero (0) initial conditions, find the Laplace transform of the following differential equation

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = 5$$

$$\left(s^2Y(s) - sy(0) - \frac{dy}{dt}(0)\right) + 3(sY(s) - y(0)) + 2Y(s) = \frac{5}{s}$$

Given zero initial conditions, $y(0) = 0$ and $\frac{dy}{dt}(0) = 0.$

Hence, the Laplace transform becomes

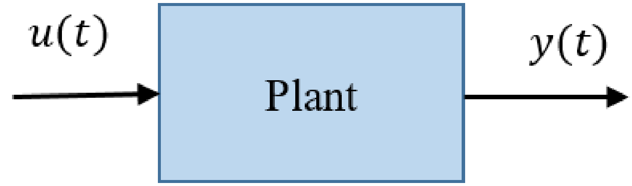
$$s^2Y(s) + 3sY(s) + 2Y(s) = \frac{5}{s}$$

Or

$$Y(s) = \frac{5}{s(s^2 + 3s + 2)}$$

# Transfer Functions

- Given a system with input $u(t)$ and output $y(t)$



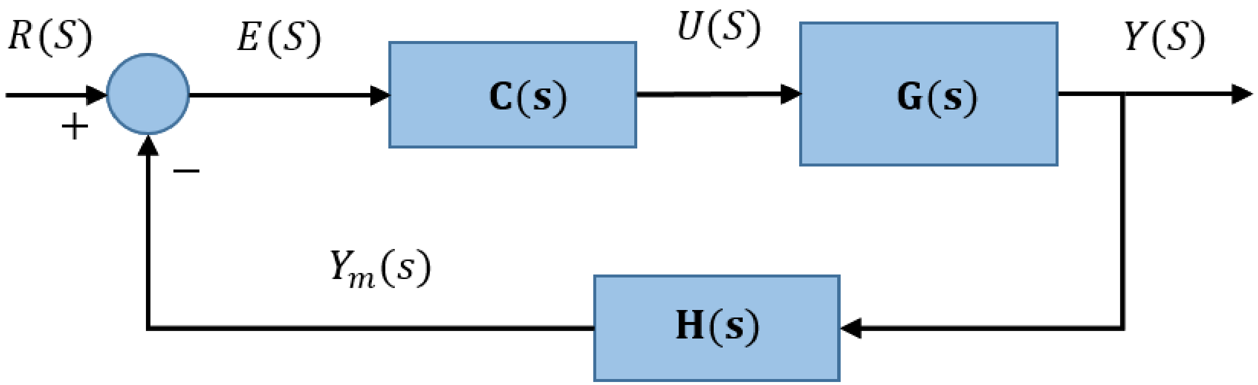

- Its transfer function is defined as

$$G(s) = \frac{Laplace\ transform\ of\ the\ input\ u(t)}{Laplace\ transform\ of\ the\ output\ y(t)}$$

- Denoting $Y(s)$ and $U(s)$ as the Laplace transforms of $y(t)$ and $u(t)$, respectively the transfer function of the plant is $G(s) = \frac{Y(s)}{U(s)}$
- The poles of $G$ are the roots of $Y(s)$ while the zeros of G are the roots of $U(s)$.
- The transfer function It is a versatile tool for studying closed-loop systems (systems with controllers).

# The Closed-loop Transfer Function

- Consider the closed-loop system




a) Plant: $\{U(s), Y(s)\}$ with $G(s) = Y(s)/U(s)$
b) Controller: $\{E(s), U(s)\}$ with $C(s) = U(s)/E(s)$
c) Sensor: $\{Y(s), Y_m(s)\}$ with $H(s) = Y_m(s)/Y(s)$

$E(s) = R(s) - Y_m(s)$

$E(s) = R(s) - Y(s)H(s)$

but $Y(s) = U(s)G(s) = C(S)G(s)[R(s) - Y(s)H(s)]$

$Y(s)[1 + C(s)G(s)H(s)] = C(s)G(s)R(s)$

Thus

$$\boxed{\boldsymbol{G_{cl} = \frac{Y(s)}{R(s)} = \frac{C(s)G(s)}{1 + C(s)G(s)H(s)}}}$$